Electronic and Telecommunication Engineering

Section 6

# Nuclear Medicine Imaging

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# Introduction

- In nuclear medicine imaging
  - A small amount (typically nanogrammes) of a radioactive material called a radiotracer is injected into the patient
  - The agent then accumulates in specific organs in the body
  - How much, how rapidly and where this uptake occurs are factors which can determine
    - Whether tissue is healthy or diseased
    - The presence of abnormalities such as tumours

# Introduction cont.

- Types of nuclear medicine imaging
  - Planar scintigraphy
    - Images the distribution of radioactive material in a single 2D image
  - Single Photon Emission Computed Tomography (SPECT)
    - Produces a series of contiguous 2D images of the distribution of the radiotracer
  - Positron Emission Tomography (PET)
    - Involves injection of a radiotracer, which emits positrons (positively charged electrons)
    - Has the highest sensitivity of the three techniques

# Introduction cont.

- Nuclear medicine scans (in particular planar scintigraphy and SPECT) have
  - Poor SNR
  - Low spatial resolution (~5–10 mm)
  - Slow image acquisition

- Provides functional information

# Introduction cont.

- Technetium-99m ($^{99m}$Tc)
  - A metastable nuclear isomer of technetium-99
  - The most common radiotracer
  - Emits mono-energetic $\gamma$-rays at 140 keV
  - Various chemical complexes of $^{99m}$Tc have been designed to target different organs in the body
  - Used in both planar scintigraphy and SPECT

# Radioactivity and Radiotracer Half-life

- A radioactive isotope
  - Undergoes a spontaneous change in the composition of the nucleus ('disintegration')
  - Emits energy
- Radioactivity (Q)
  - The number of disintegrations per second, or alternatively the time rate of change of the number of radioactive nuclei
  - SI unit: Becquerel (Bq)
  - 1 Ci $= 3.7 \times 10^{10}$ disintegrations per second
  - For $N$ nuclei of a particular radioactive isotope:

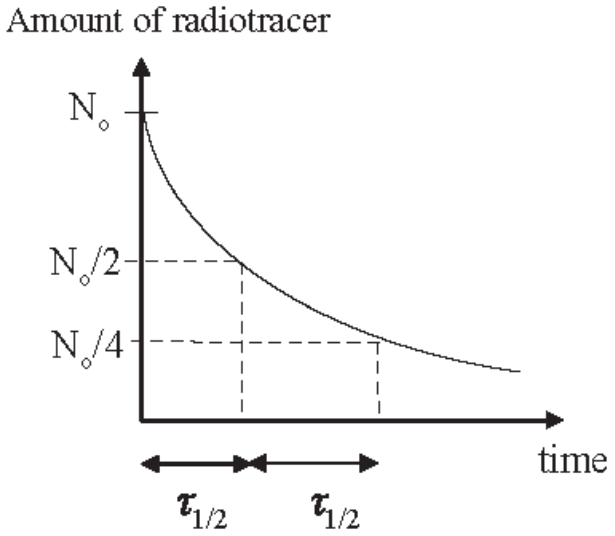
$$Q = -\frac{dN}{dt} = \lambda N$$

where $\lambda =$ decay constant (units: $s^{-1}$)

# Radioactivity and Radiotracer Half-life cont.

- The number of nuclei at time $t$:

$$N(t) = N(t=0)e^{-\lambda t}$$




- Half-life ($\tau_{1/2}$)
  - The time required for the radioactivity to drop to one-half of its value
  - The most common measure of radioactivity
  - Independent of the value of $N$

$$\tau_{1/2} = \frac{\ln 2}{\lambda}$$

# Radioactivity and Radiotracer Half-life cont.

- Biological half-life of the radiotracer ($\tau_{1/2,\text{bio}}$)
  - Determines how long the radiotracer remains in the body
  - Excretion of the radiotracer from tissue is an exponential decay

- The effective half-life ($\tau_{1/2,\text{eff}}$)
  - Combination of the two half-lives, $\tau_{1/2}$ and $\tau_{1/2,\text{bio}}$
  - Always less than the shorter of the two half-lives

$$\tau_{1/2,\text{eff}} = \frac{\tau_{1/2} \cdot \tau_{1/2,\text{bio}}}{\tau_{1/2} + \tau_{1/2,\text{bio}}}$$

# Ideal Properties of Radiotracers

For planar scintigraphy and SPECT:

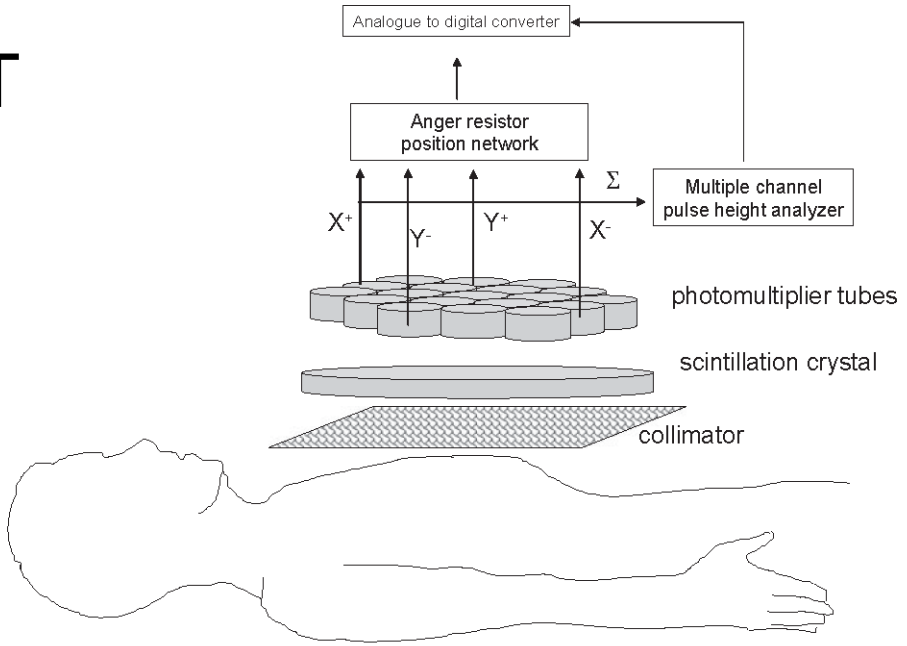
- Radioactive half-life
  - Short enough to produce significant radioactivity without requiring a very large initial dose
  - No significant decay before the required post-injection delay to allow the radiotracer to clear the blood and distribute in the relevant organs
- Emission during decay
  - Mono-energetic $\gamma$-rays without alpha or beta particles
  - A mono-energetic $\gamma$-ray allows discrimination between Compton scattered and unscattered $\gamma$-rays
  - Alpha or beta particles are completely absorbed within tissue increasing the radioactive dose without giving any useful image information

# Ideal Properties of Radiotracers cont.

- The energy of the $\gamma$-ray
  - Should be greater than ~100 keV for a reasonable proportion of $\gamma$-rays emitted deep within the tissue have sufficient energy to travel through the body and reach the detector
  - Should be less than ~200 keV to prevent the rays penetrating the thin lead septa in the collimator
- Tracer uptake
  - Should be high in the organ of interest
  - Should be relatively low in the rest of the body
  - Above two factors lower the required dose for the patient and increase the image contrast

# The Gamma Camera

- The instrumental basis for both planar scintigraphy and SPECT
  - Must be capable of $\gamma$-ray detection rates of up to tens of thousands per second
  - Should reject those $\gamma$-rays that have been scattered
  - Must have a high sensitivity

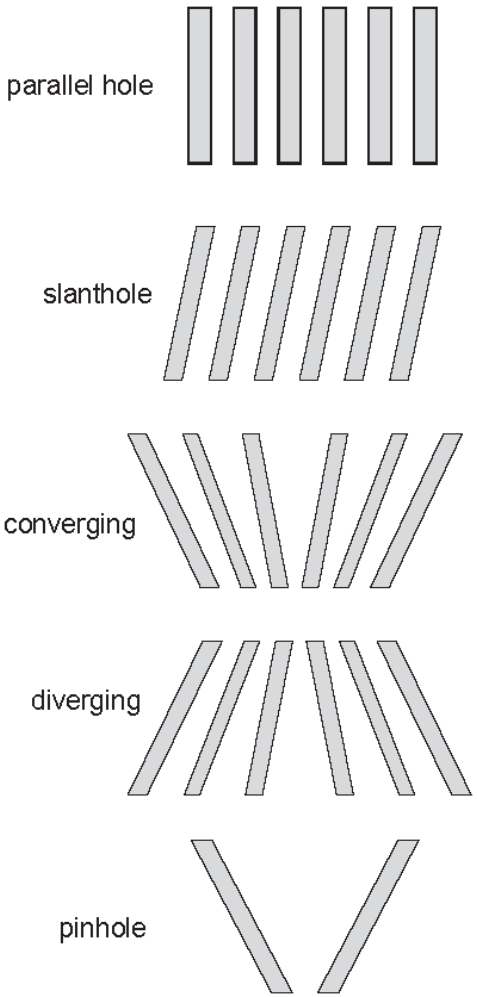


# The Gamma Camera cont.

- The collimator
  - Very similar to role of the anti-scatter grid in X-ray imaging
  - Higher degree of collimation is required since $\gamma$-rays from a source of radioactivity within the body are emitted in all directions
  - Leads to a very high proportion of emitted $\gamma$-rays (>99.9%) not being detected (together with the high attenuation of $\gamma$-rays within the body)

# The Gamma Camera cont.

- Collimator designs
  - Parallel hole:
    - Most common
  - Slanthole:
    - For breast and cardiac imaging
  - Converging:
    - To magnify the image and increase the spatial resolution
  - Diverging:
    - Enables a larger FOV
    - Example: planar scintigraphy of the whole body
  - Fan-beam:
    - Two dimensions of the collimator have different geometries: parallel hole and converging
    - Provides image magnification over a reduced FOV
    - For brain and heart studies
  - Pinhole:
    - Produces an image with significant magnification and higher spatial resolution
    - For imaging very small organs such as the thyroid and parathyroid



# Instrument Dead Time

- Number of $\gamma$-rays striking the scintillation crystal exceeding the recording capabilities
  - When an injected dose of radiotracer is very large (usually at the beginning of a clinical scan)
  - Due to the finite recovery and reset times required for various electronic circuits in the gamma camera

- Two types of behaviour exhibited by the system components
  - Paralysable
  - Non-paralysable

# Instrument Dead Time cont.

- Paralyzable
  - System cannot respond to a new event until a fixed time after the previous one, irrespective of whether that component is already in a non-responsive state
  - Non-responsive time is elongated, only one set of photons will be produced

- Non-paralyzable
  - Components cannot respond for a set time, irrespective of the level of radioactivity
  - Any further events are simply not recorded: this time is fixed, and is not elongated

# Instrument Dead Time cont.

- The overall 'dead-time' ($\tau$)
  - Usually determined by the non-paralyzable components

$$\tau = \frac{N - n}{nN}$$

Where;

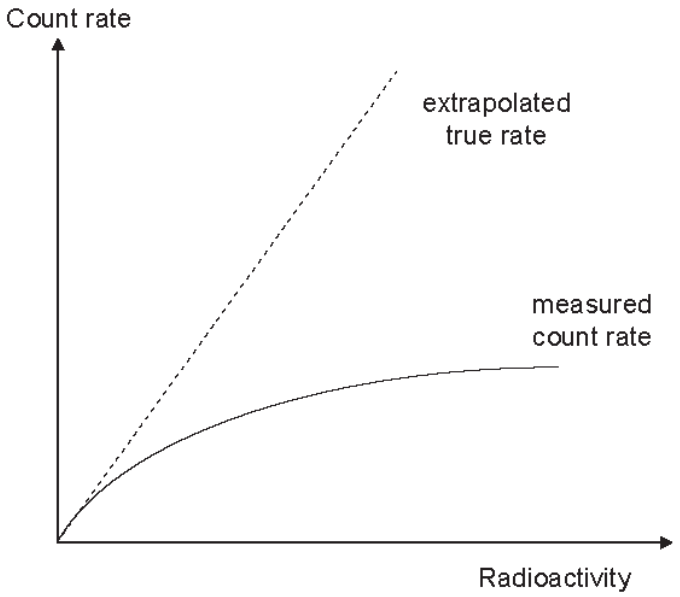
$N =$ true count rate (number of scintillations per sec.)

$n =$ measured count rate

The maximum measurable count rate $= 1/\tau$

# Instrument Dead Time cont.

- Corrections for the dead-time
  - Based upon calibrations using phantoms of known radioactivity
  - Measurements of the dead time of individual components of the system
  - Calibration curve: for correcting the acquired images



# Clinical Applications of Planar Scintigraphy

- The major clinical applications
  - Whole-body bone scanning
  - Thyroid, gastrointestinal tract, liver and kidney scans with specialized agents

- Diagnosis of pathological conditions such as cirrhosis of the liver
  - $^{99m}$Tc attached to small particles of sulphur colloid, which concentrate in healthy Kupffer cells in the liver, spleen and bone marrow
  - 'cold spots' in areas of diseased cells due to no radiotracer uptake

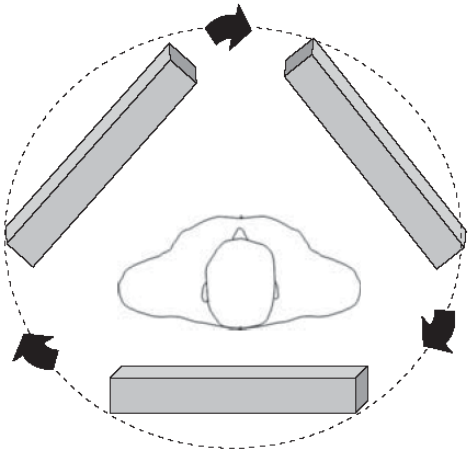
# Clinical Applications of Planar Scintigraphy cont.

- Bone scanning and tumour detection
  - Using $^{99m}$Tc phosphonates such a hydroxymethylene-diphosphonate (HMDP)
  - Detects bone tumours and soft-tissue tumours which cause deformation and remodeling of bone structure
  - Scanning starts 2–3 hours after injection of the radiotracer into a patient's bloodstream to allow accumulation within the skeletal structure



# Single Photon Emission Computed Tomography (SPECT)

- Uses two or three gamma cameras which rotate around the patient detecting $\gamma$-rays at a number of different angles
- Either filtered back-projection or iterative techniques are used to reconstruct multiple two-dimensional axial slices
- Uses essentially the same instrumentation and many of the same radiotracers as planar scintigraphy





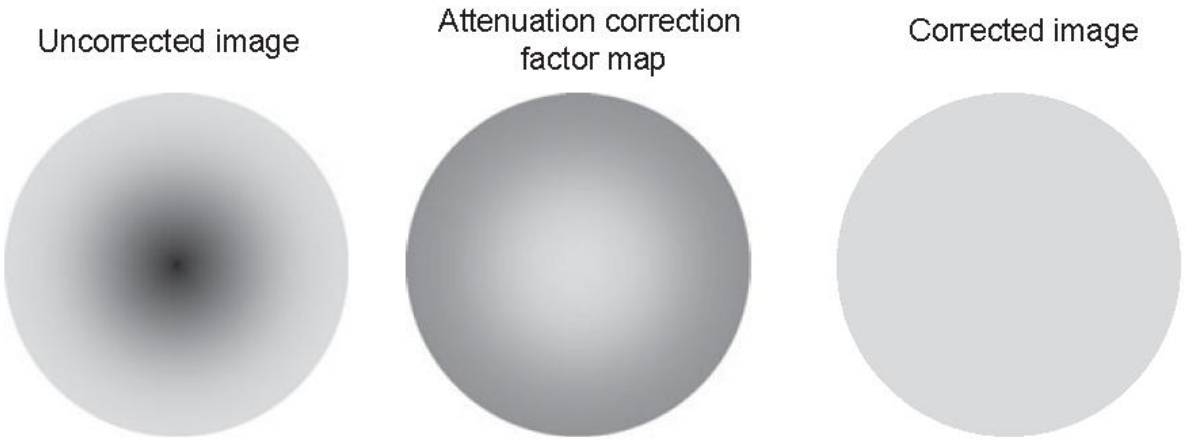
# Single Photon Emission Computed Tomography (SPECT) cont.

- Major applications
  - Myocardial perfusion studies to detect coronary artery disease or myocardial infarction
  - Brain studies to detect areas of reduced blood flow associated with stroke, epilepsy or neurodegenerative diseases such as Alzheimer

- Scatter correction
  - The first step in data processing
  - The contribution from scattered $\gamma$-rays is typically of the same order as those which have not been scattered
  - Uses a dual-energy, or multiple-energy window and a multi-channel analyzer

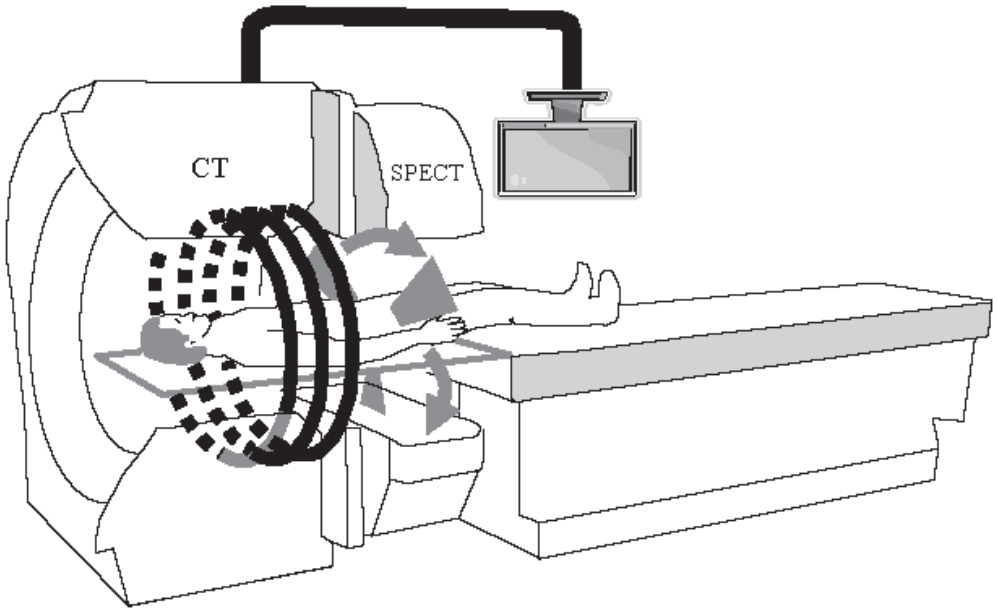
# Single Photon Emission Computed Tomography (SPECT) cont.

- Attenuation correction
  - Required for accurate quantitation since $\gamma$-rays from deeper within the body are attenuated more
  - Two main techniques
    1. Chang's multiplicative method assumes that the tissue attenuation coefficient is uniform and uses pre-calculated attenuation correction factor map to correct the attenuation
    2. Using a transmission scan, usually acquired simultaneously with the actual patient scan with a source containing radioactive gadolinium (Gd) with low energy $\gamma$-rays and longer half-life



# SPECT/CT Systems

- Acquires data from the two image modalities with a single integrated patient bed and gantry
- Typical clinical systems use a two-head SPECT camera with a multi-slice CT scanner



# SPECT/CT Systems cont.

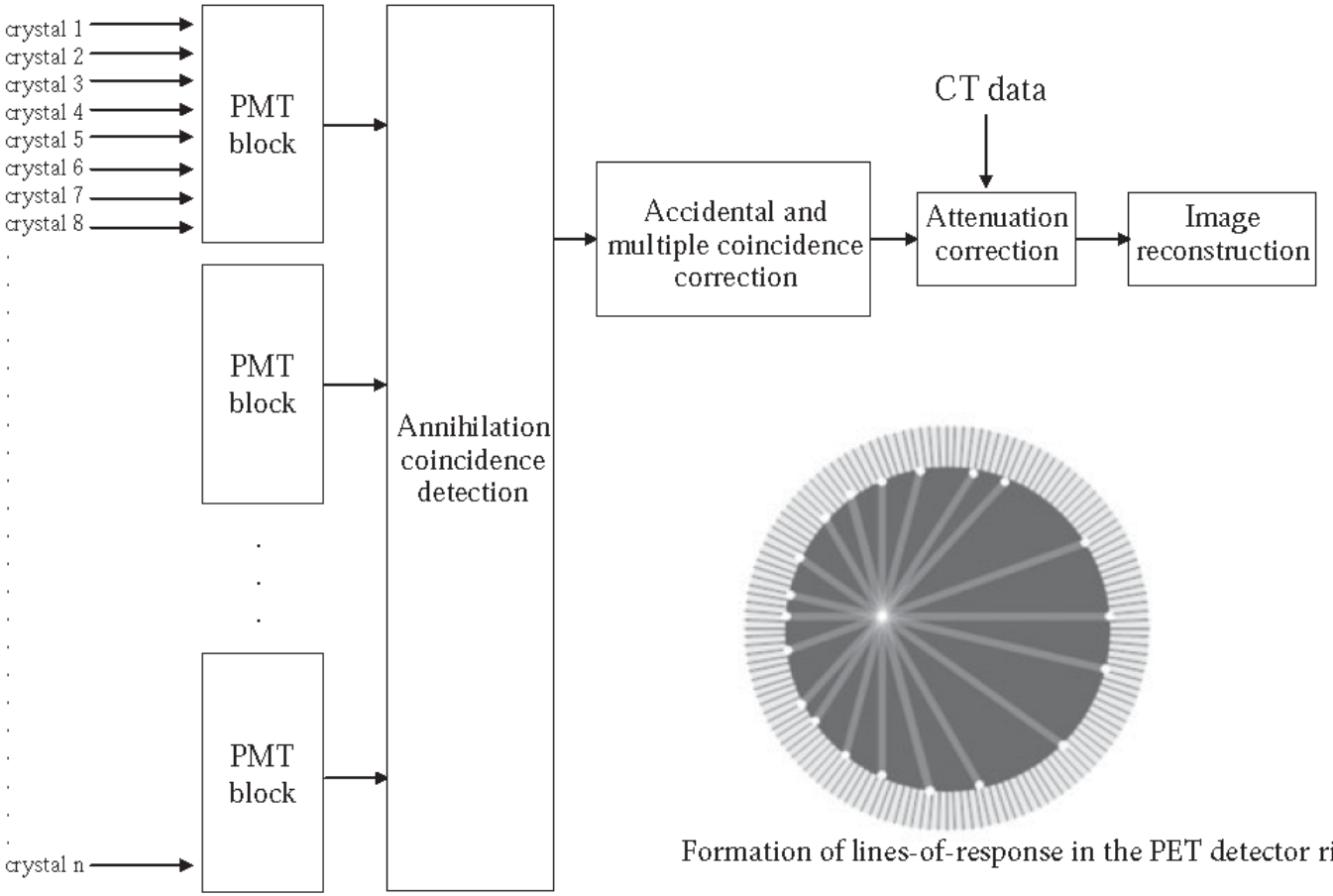
- Advantages of a combined SPECT/CT over a stand-alone SPECT
  - Improved attenuation correction for SPECT reconstruction using high-resolution anatomical information from the CT scanner
  - Fusion of high-resolution anatomical (CT) with functional (SPECT) information

# Positron Emission Tomography (PET)

- A tomographic technique which also uses radiotracers similar to SPECT
- Differences of PET over SPECT
  - PET has between 100 and 1000 times higher SNR
  - PET has a significantly better spatial resolution
- Radiotracers used in PET emit positrons
  - Produce two $\gamma$-rays with energies of 511 keV at $180^{o}$ apart after annihilation with an electron in tissue detected by a complete ring of detectors
  - Forms an intrinsic line-of-reconstruction (LOR) without the need for any collimation

# Positron Emission Tomography (PET) cont.

- Typical PET system




Formation of lines-of-response in the PET detector ring.

# Positron Emission Tomography (PET) cont.

- Factors for higher SNR of PET compared to SPECT
  - Collimation not being required
  - Reduced attenuation of higher energy (511 keV vs. 140 keV) $\gamma$-rays in tissue
  - Use of a complete ring of detectors
- Hybrid PET/CT scanners
  - Use a single patient bed to slide between the two systems
  - Has improved attenuation correction
  - Capable of fusing morphological and functional information

# Positron Emission Tomography (PET) cont.

- Major clinical applications
  - General areas of oncology, cardiology and neurology
  - In oncology, whole body imaging is used to identify both primary tumours and secondary metastatic disease remote from the primary source



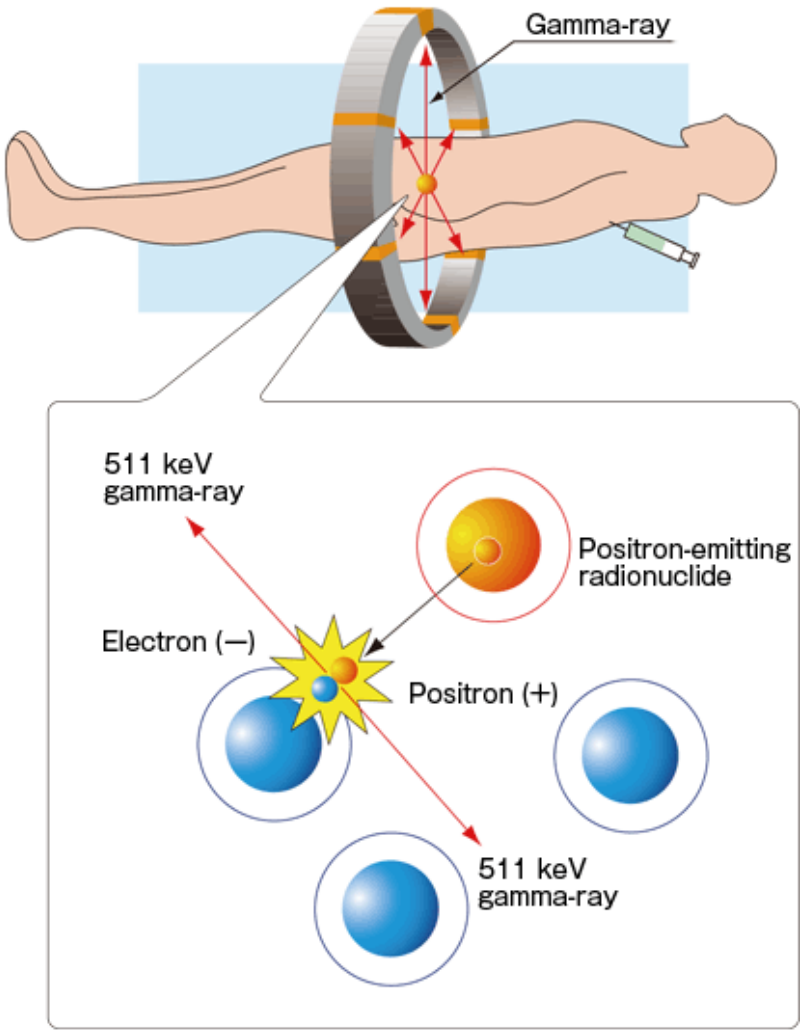
- Major disadvantages
  - Requirement for an on-site cyclotron to produce positron emitting radiotracers, and the high associated costs

# Positron Emission Tomography (PET) cont.

- Radiotracers used for PET
  - Isotopes such as $^{11}C$, $^{15}O$, $^{18}F$ and $^{13}N$
  - Undergo radioactive decay by emitting a positron, i.e. a positively charged electron ($e^+$), and a neutrino
  - Positron travels a short distance (an average of 0.1–2 mm depending upon the particular radiotracer) in tissue before annihilating with an electron
  - Annihilation results in the formation of two $\gamma$-rays, each with an energy of 511 keV at $180^o$ apart
  - Generally, have a short half-life

# Positron Emission Tomography (PET) cont.

- Radioactive tracer isotope is injected
  - FDG (Fluorodeoxyglucose)
- Radioisotope emits positrons
- Electron-positron annihilation
  - Two 511 keV photons travelling in opposite directions
- Image of tracer concentration is reconstructed by coincident detections of annihilation photon pairs




*Source:https://www.hamamatsu.com/us/en/technology/innovation/pet/index.html*

# Positron Emission Tomography (PET) cont.

- Radiotracers used for PET: properties

| Radionuclide | Half-life (minutes) | Radiotracer | Clinical applications |
|---|---|---|---|
| $^{18}F$ | 109.7 | $^{18}FDG$ | oncology, inflammation, cardiac viability |
| $^{11}C$ | 20.4 | $^{11}C$-palmitate | cardiac metabolism |
| $^{15}O$ | 2.07 | $H_2{}^{15}O$ | cerebral blood flow |
| $^{13}N$ | 9.96 | $^{13}NH_3$ | cardiac blood flow |
| $^{82}Rb$ | 1.27 | $^{82}RbCl_2$ | cardiac perfusion |

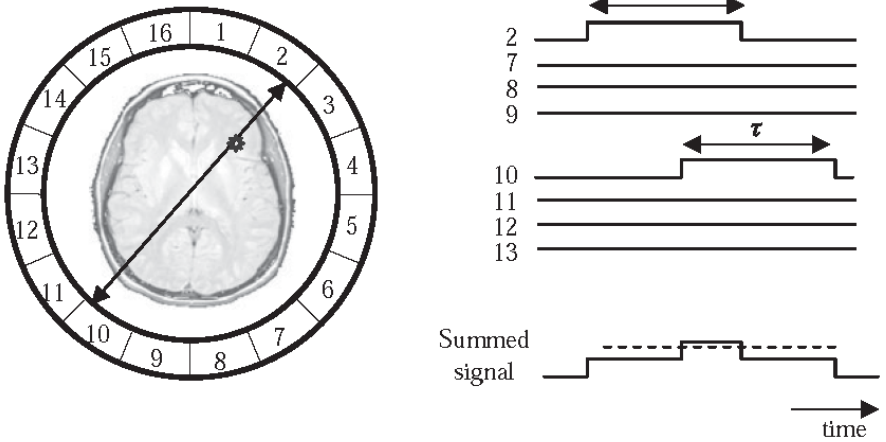
- Instrumentation for PET/CT
  - Many thousands of small solid-state detectors
  - Annihilation coincidence circuitry
  - Multi-channel pulse height analyzers and photomultiplier tubes similar to SPECT

# Positron Emission Tomography (PET) cont.

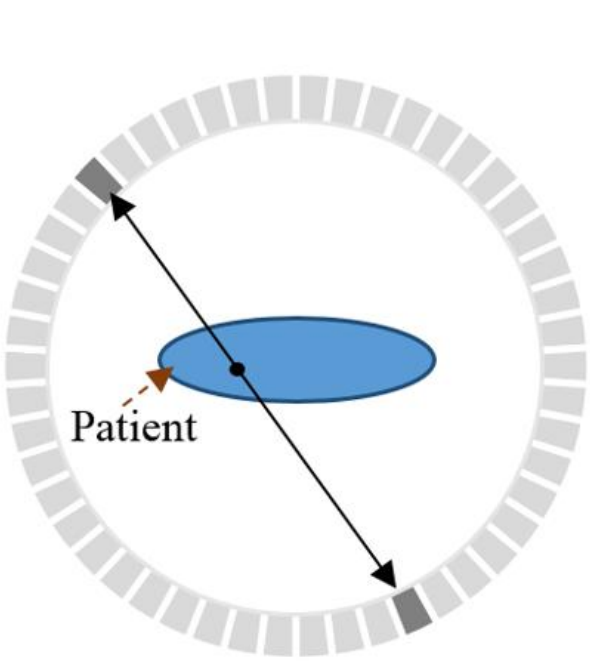
- Annihilation Coincidence Detection (ACD)
  - A fixed ‘coincidence resolving time’ is defined as the time-window that is allowed after the first $\gamma$-ray has been detected for a second $\gamma$-ray to be recorded and assigned to the same annihilation
  - A pair of $\gamma$-rays is accepted as a ‘true coincidence’ if second $\gamma$-ray strikes a detector which is operated in coincidence with the first detector and an LOR can be drawn between the two detectors

# Positron Emission Tomography (PET) cont.

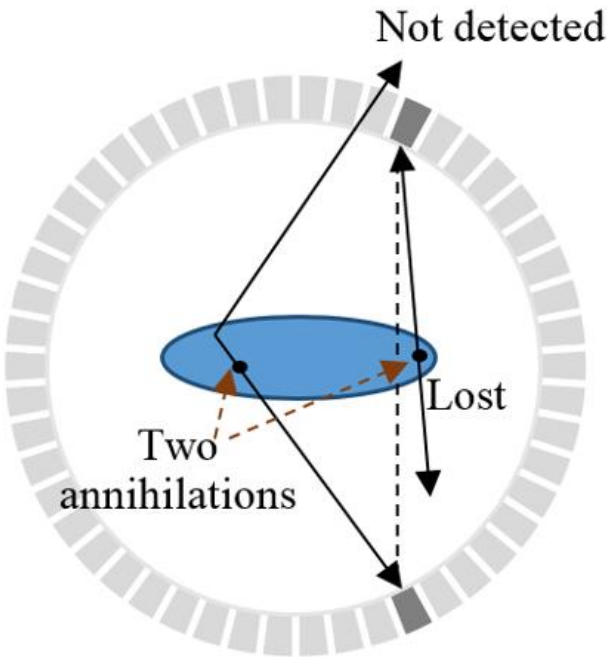
- ACD Example
  - The first $\gamma$-ray from the annihilation strikes detector no. 2
  - Only those detectors numbered between 7 and 14 are 'in coincidence' with detector no. 2 since unscattered $\gamma$-rays originate within the head
  - The second $\gamma$-ray is detected in crystal 10 is added to the first pulse
  - If the summed signal is greater than a threshold set to a value just less than twice the amplitude of each individual logic pulse, then a coincidence is accepted, and an LOR established between the two detectors.
  - The coincidence resolving time is defined as twice the length of each logic pulse



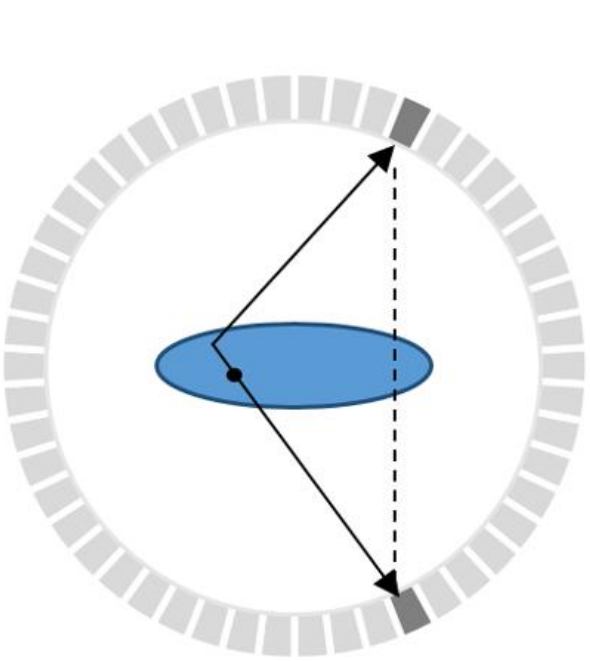
# Types of Coincidences




True coincidence



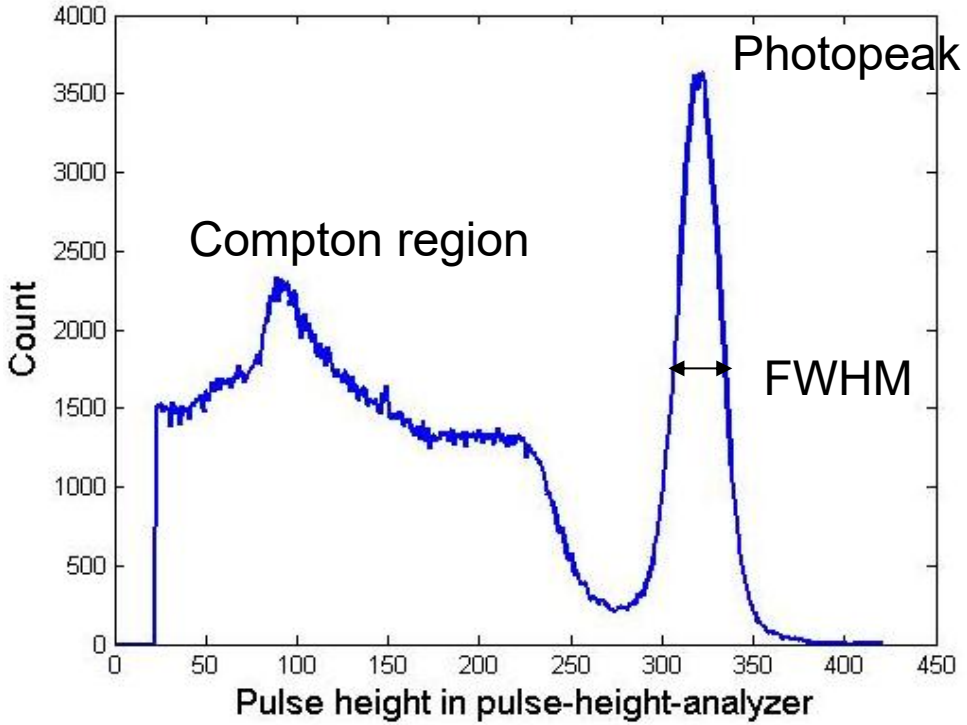

Random coincidence



Scattered coincidence

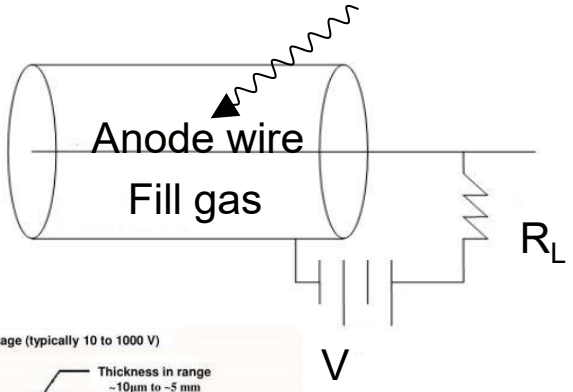
# Radiation Detectors

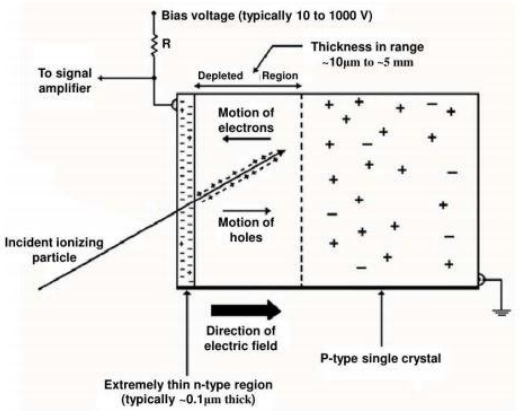
- Energy deposited by radiation → measurable electrical signal or charge

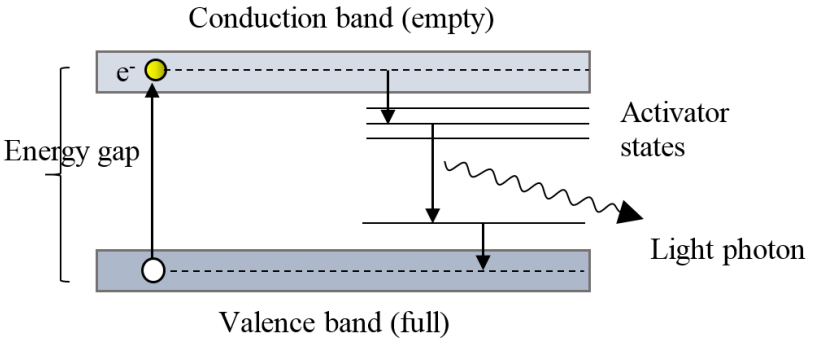


# Radiation Detectors cont.

- Proportional gas chambers
  - Low stopping efficiency for 511 keV photons
  - Poor energy resolution
- Semiconductor detectors
  - Excellent energy resolution
  - Stopping efficiency for 511 keV photons is low
- Scintillation detectors
  - Highest stopping efficiency for 511 keV photons
  - Moderate energy resolution

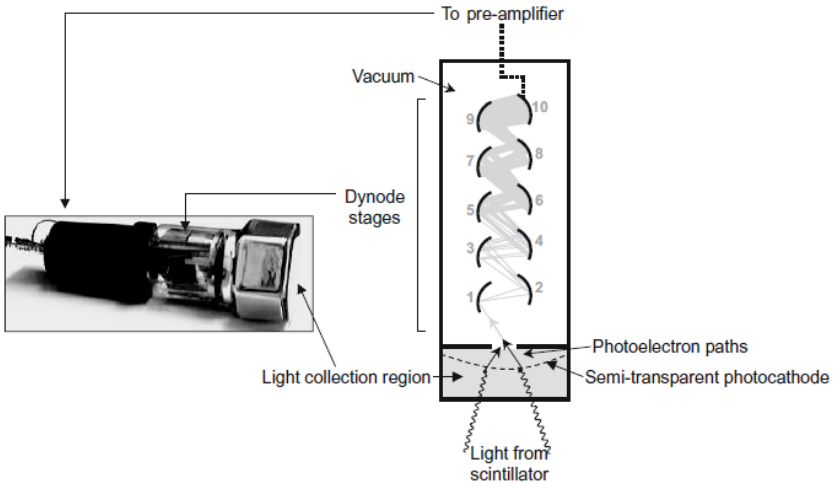


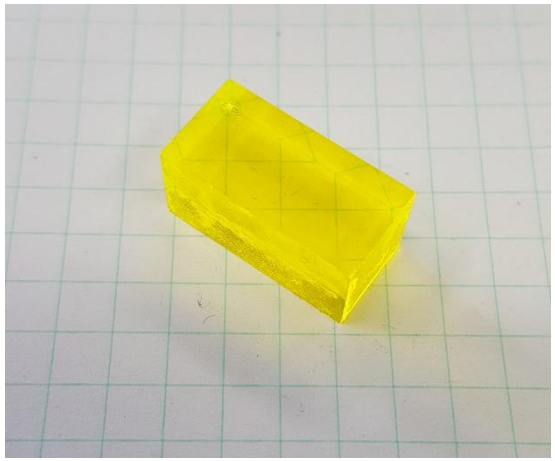
# Radiation Detectors: Photodetectors

1. Photomultiplier tube
   - High gain of more than $10^6$
   - Very good SNR
   - Operating voltage of 1000 to 2000 V
   - Limited by their sizes and sensitivity to magnetic fields

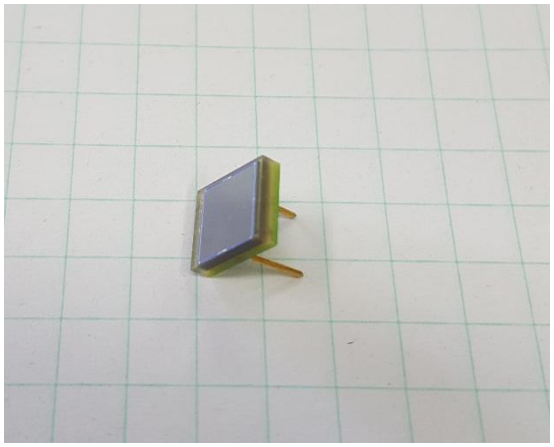



2. Semiconductor based devices
   - PIN diodes
     - Low SNR
   - Avalanche photo diode (APD)
     - Improved SNR
     - Gain of few hundreds
   - Silicon photomultiplier
     - A matrix of Geiger-mode avalanche photodiodes
     - Compact
     - Gain is approximately $10^6$
     - Low operating voltage

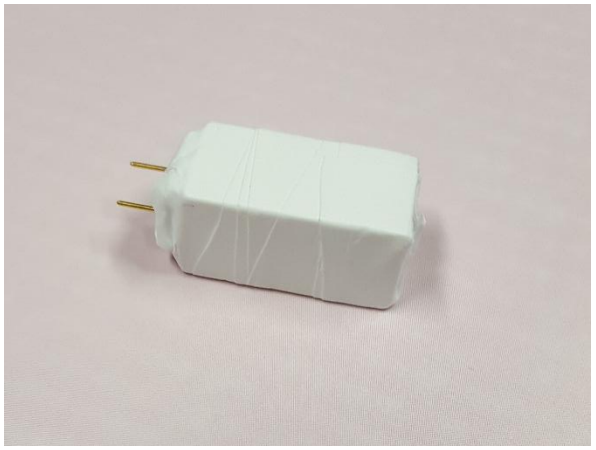
# Radiation Detectors cont.



GAGG crystal



SiPM



GAGG coupled with SiPM



# Positron Emission Tomography (PET)

- 2D and 3D PET imaging
  - Number of detector rings up to 48 or more stacked along the patient head/foot direction
  - Data can be acquired either in 2D multi-slice or full 3D mode
  - Retractable lead (Pb) collimation septa positioned between each ring
    - Extended for 2D multi-slice operation and retracted for imaging in 3D mode
    - Reduces the amount of scattered $\gamma$-rays detected in 2D mode
    - Produces a uniform sensitivity profile along the axial dimension

# Positron Emission Tomography (PET)

- 2D and 3D PET imaging
  - 3D PET has a much higher sensitivity than 2D PET by about a factor of 10, primarily due to the elimination of the collimators
  - Three different data acquisition modes used for PET scans (septal collimators are retracted for 3D mode)

